

Bill of Materials — 2026-2027 MATE ROV Season (RANGER class)

Madison, Alabama · Updated June 2026 · Maintained by the CFO

Category	Total budget	Funded	Remaining	Status
Tier 1 — Competition entry & travel	\$1,650	\$1,650	\$0	Funded
Tier 2 — Propulsion, frame & power	\$2,800	\$1,050	\$1,750	Partial
Tier 3 — Electronics, control & payload	\$2,320	\$300	\$2,020	Partial
Tier 4 — Tools, safety & shop	\$1,400	\$0	\$1,400	Open
Tier 5 — Float, documentation & reliability	\$1,630	\$0	\$1,630	Open
Season total	\$9,800	\$3,000	\$6,800	31%

How to read this workbook

- One sheet per Tier. Each sheet lists every line item, vendor, quantity, unit cost, total, and funding status.
- Change the Status column on any line to “Funded”, “Ordered”, or “Received” as money lands and parts arrive.
- Summary totals roll up automatically from the Tier sheets.
- The 2027 MATE competition manual drops in November 2026 — expect mission-specific items to be added then.

Tier 1 — Competition entry & travel

Gets the team to the water in April 2027.

#	Item / Part	Vendor	Part / Reference	Qty	Unit cost	Total	Status
1	RANGER competition registration	MATE / DISL	2027 Northern Gulf Coast Regional	1	\$250	\$250	Funded
2	Regional travel & lodging	Team-managed	Dauphin Island, AL — 6 hr drive	1	\$1,400	\$1,400	Funded
Tier 1 total						\$1,650	

#	Item / Part	Vendor	Part / Reference	Qty	Unit cost	Total	Status
1	Brushless thruster (Blue Robotics T200) + ESC	Blue Robotics	T200-THRUSTER-R2 + BasicESC-R3	6	\$290	\$1,740	Open
2	Spare thruster + printed propellers	Blue Robotics + in-house	T200 spare + spare props	1	\$300	\$300	Funded
3	ROV frame & structure (extrusion, HDPE, 316SS hardware)	Misumi / McMaster	20×20 alu extrusion + plate	1	\$450	\$450	Funded
4	12 V battery system (LiFePO4 pack, charger, enclosure)	Bioenno + Powerwerx	LiFePO4 12V 12-20Ah + IP65 box	1	\$300	\$300	Funded
Tier 2 total						\$2,790	

#	Item / Part	Vendor	Part / Reference	Qty	Unit cost	Total	Status
1	Watertight electronics enclosure (4" tube, end caps, clamp, rails)	Blue Robotics	WTE4-ASM-R1	1	\$300	\$300	Open
2	Penetrators & waterproof connectors	Blue Robotics	WetLink + Cobalt 8 + spares	1	\$300	\$300	Open
3	Pre-terminated tether (~25 m, neutrally buoyant)	Blue Robotics	Fathom Tether 8-conductor	1	\$600	\$600	Open
4	Onboard compute & control (Raspberry Pi 5, Pixhawk-style FC, topside box)	Adafruit + Blue Robotics	Pi 5 8GB + Navigator	1	\$400	\$400	Open
5	Upgraded camera & vision system (low-light HD)	Blue Robotics	Low-Light HD USB Camera	1	\$250	\$250	Open
6	Depth & orientation sensors (pressure + IMU)	Blue Robotics + Adafruit	Bar30 + BNO055	1	\$170	\$170	Open
7	Pneumatic manipulator (gripper + small cylinder + tubing)	Misc / DIY	Gripper + pneumatic cylinder	1	\$300	\$300	Funded
Tier 3 total						\$2,320	

#	Item / Part	Vendor	Part / Reference	Qty	Unit cost	Total	Status
1	Soldering stations & irons (workshop set)	Hakko / Weller via Amazon	FX-888D x 2	2	\$150	\$300	Open
2	Hand tools & test equipment (multimeters, clamp meter, etc.)	Amazon + McMaster	Fluke 117 + clamp meter kit	1	\$400	\$400	Open
3	Safety gear (fume extractor, fire extinguisher, first-aid, mats)	Various	Workshop safety kit	1	\$180	\$180	Open
4	Adhesives, sealants & O-ring supplies	Loctite + McMaster	Marine epoxy + grease + O-rings	1	\$200	\$200	Open
5	Build consumables (wire, solder, heat-shrink, fasteners, props)	Various	Bulk pack	1	\$200	\$200	Open
6	Soldering & circuit practice kits (every student)	Adafruit / Elenco	Practice boards (qty 12)	1	\$120	\$120	Open
Tier 4 total						\$1,400	

#	Item / Part	Vendor	Part / Reference	Qty	Unit cost	Total	Status
1	Vertical profiling float build (extrusion, buoyancy engine, electronics)	Blue Robotics + 3D print	Custom build per MATE Floats spec	1	\$400	\$400	Open
2	Float pressure sensor + battery + radio	Adafruit + Blue Robotics	Bar02 + 7.4V LiPo + LoRa radio	1	\$230	\$230	Open
3	Print + bind the technical documentation + marketing display	Local print shop	Color print + binding + foam-core	1	\$250	\$250	Open
4	Engineering presentation tablet + display dongle	Used iPad / Amazon	Display kit for the judges' room	1	\$250	\$250	Open
5	Practice ball / mission props (community-pool rehearsal aids)	DIY	Misc PVC / 3D print	1	\$200	\$200	Open
6	Photogrammetry license + storage (RealityCapture or Meshroom)	Capturing Reality	Steam license per-input	1	\$300	\$300	Open
Tier 5 total						\$1,630	